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Subject : Virtualization and Cloud Computing

Prof. : Sumit Kalra

Assignment 1:

Using VirtualBox to Create Multiple VMs, Connect Them, and Host a Microservice-Based Application

Introduction:

The assignment is to create and configure multiple Virtual Machines (VMs) using Oracle VirtualBox, establish network connectivity between them, and deploy a microservice-based application across the connected VMs.

Implementation Process:

1. Installation of VirtualBox and Creation of Multiple VMs:

- Download and Install Oracle VirtualBox from the official website. and I downloaded **Ubuntu ISO 22.04 ISO** from the official Ubuntu website.
- Then I created Two Virtual Machines (VMs) by assigning names as VM Master and VM Nodes with "Ubuntu (64-bit).iso". The following are the VMS configurations.
    - For each VM **1.7 - 2.5 GB RAM** and **30 GB ROM**.
    - And with **2 CPU cores** in each VM.
    - Booted both VMs and performed initial setup.

2. Configuration of Network Settings to Connect the VMs

- I checked the Network Configuration by using **“ifconfig”** and **“ip a”** to check IP addresses of both VMs. Initially, both VMs had the same IP address.
- To Change Network Mode, in VirtualBox settings for each VM, I changed the Network Settings of the adapter mode from **NAT** to **Bridged Adapter**.
- And I restarted both Virtual Machines to apply **Bridged Adapter** network changes.
- For assigning Unique IP Addresses, I run the following command in each VMs **“dhclient -r && dhclient”**
- Now both VMs have their own separate IPS, which is suitable for making connections between them.
- Installation and Enable SSH by updating the following package lists: **“sudo apt update”** and **“sudo apt install openssh-server”**
- And to Enable and start the SSH service by **“sudo systemctl enable ssh”** and **“sudo systemctl start ssh”**
- To verify SSH status I used **“systemctl status ssh”,** which successfully resulted as active.
- And to Test Network Connectivity, I used the ping command to check the connection between VMs **“ping 172.20.10.10“(ip address for Vm Node1)**
- Packet exchange confirms network establishment is successful.

3. Deployment of a Simple Microservice Application

- I selected “**News Aggregator”** for Microservice Application to communicate between VMs.
- I started by installing the required packages such as Python and pip3 using the command:

  “sudo apt install python3 python3-pip”
- And I Installed Flask and Requests by **“pip3 install flask requests”.**

- I used **“News API Key”** from newsapi.org and I stored API key in an environment variable by adding the following line to ~/.bashrc:

  “export NEWS_API_KEY='6441d65sasf2sdf6fzf'” (#pseudo API key)

- And to apply changes: **“source ~/.bashrc”**
- I created Flask Application by writing the Python script named **“fetch.py”.**

```
from flask import Flask, jsonify
import requests
import os
app = Flask(__name__)
@app.route('/')
def get_news():
api_key = os.getenv("NEWS_API_KEY")
url = f'https://newsapi.org/v2/top-headlines?country=us&apiKey={api_key}'
response = requests.get(url)
return jsonify(response.json())
if __name__ == '__main__':
app.run(host='0.0.0.0', port=5000)
```

- Now to run the Microservice using “**python3 fetch.py”**
- I tested the Microservice Locally by using **“curl http://localhost:5000/news”**
- This resulted successfully, by returning the latest news headlines.

4. Running the Microservice Across the VM

- For accessing the Microservice from VM2, I Connected to VM1 from VM2 using SSH

  “ssh vboxuser@172.20.10.9”

- Once the VMs are connected I ran **“fetch.py”** and established a flask server alive. In another terminal tab, I SSHed with VM1 from VM2 and ran **“curl http://localhost:5000/news”.** This resulted in displaying live news from the microservice running on VM1 in VM2.

Deploying Code to GitHub

- Initialized Git Repository by **“git init”** and Added Files and Commit Changes by

  “git add .”

  “git commit -m "Initial commit"”

- To push Code to GitHub, **“git remote add origin <GitHub-Repo-URL>”**

  “git branch -M main”

  “git push -u origin main”

Architecture Design

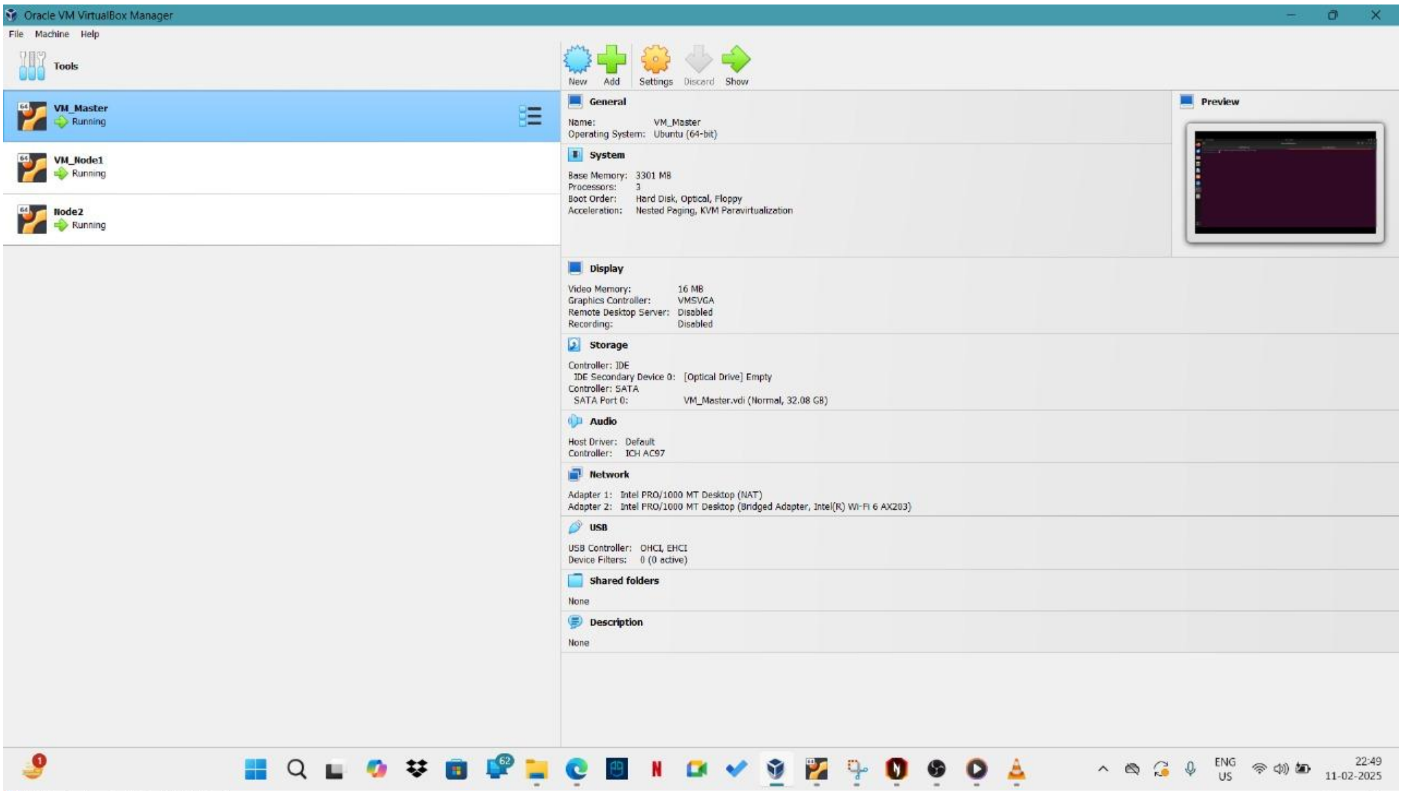
```
        +-----------------+            +-----------------+
     |  Virtual Machine 1  |        |  Virtual Machine 2  |
   | (Ubuntu, Flask App) |  <--->  | (Ubuntu, SSH Client) |
     |   IP: 172.20.10.9  |           |   IP: 172.20.10.10  |
     +-----------------+                  +-----------------+
           |                                      |
           |-------->     Bridged  Network     <------|
                          |          |
          +-------------------------------------------+
       |          Oracle VirtualBox Host Machine        |
          +-------------------------------------------+
```

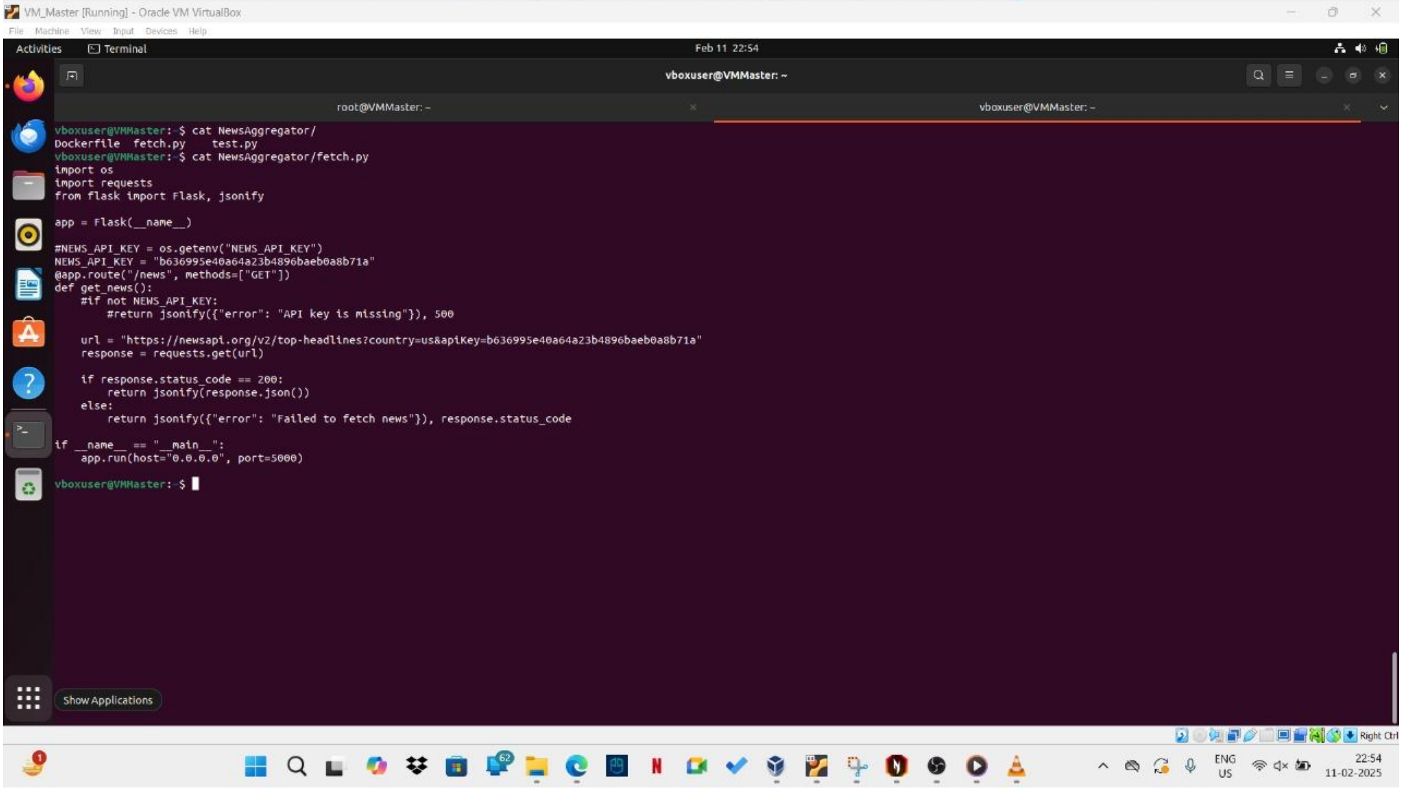
Link to Source Code Repository and Video Recording

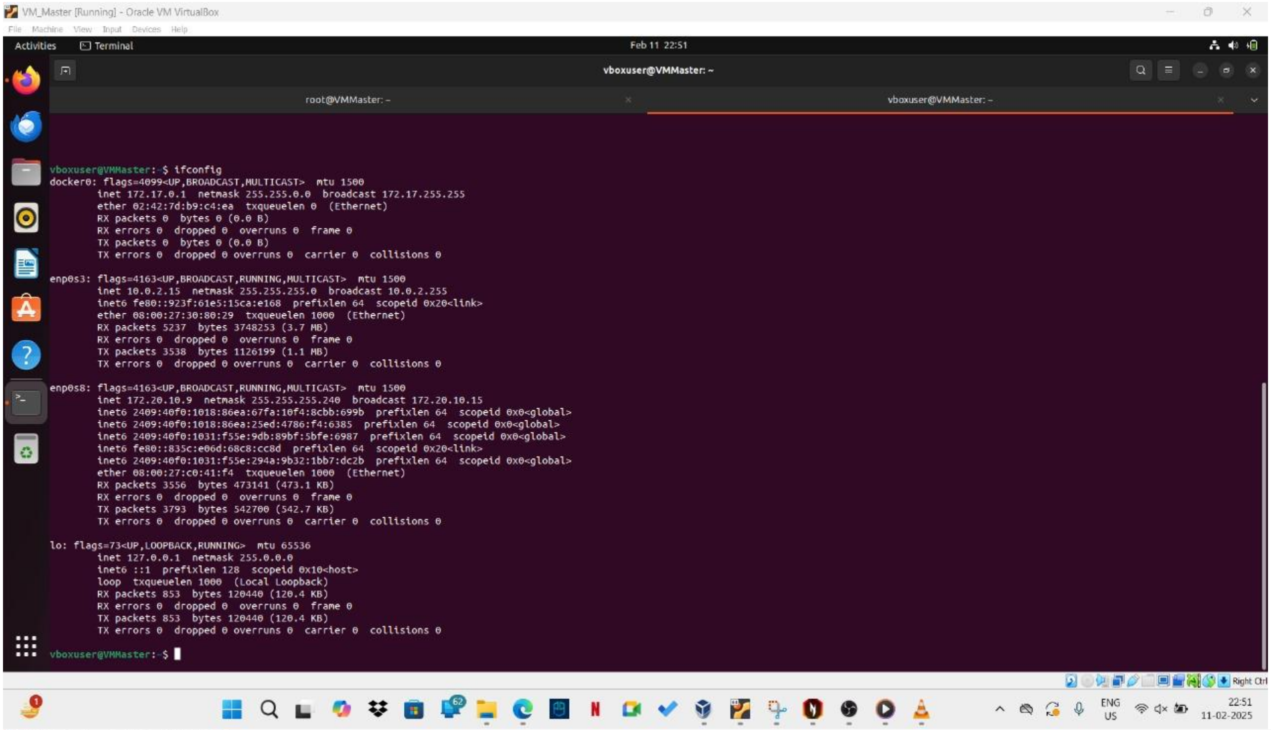
- **GitHub Repository**: https://github.com/saiswaroopkakarla/NewsAggregator.git
- **Video Demo:** https://drive.google.com/file/d/12f2NrZ3KtkRFnNkzP8ZA3xGpvm0OBUBd/view?usp=sharing

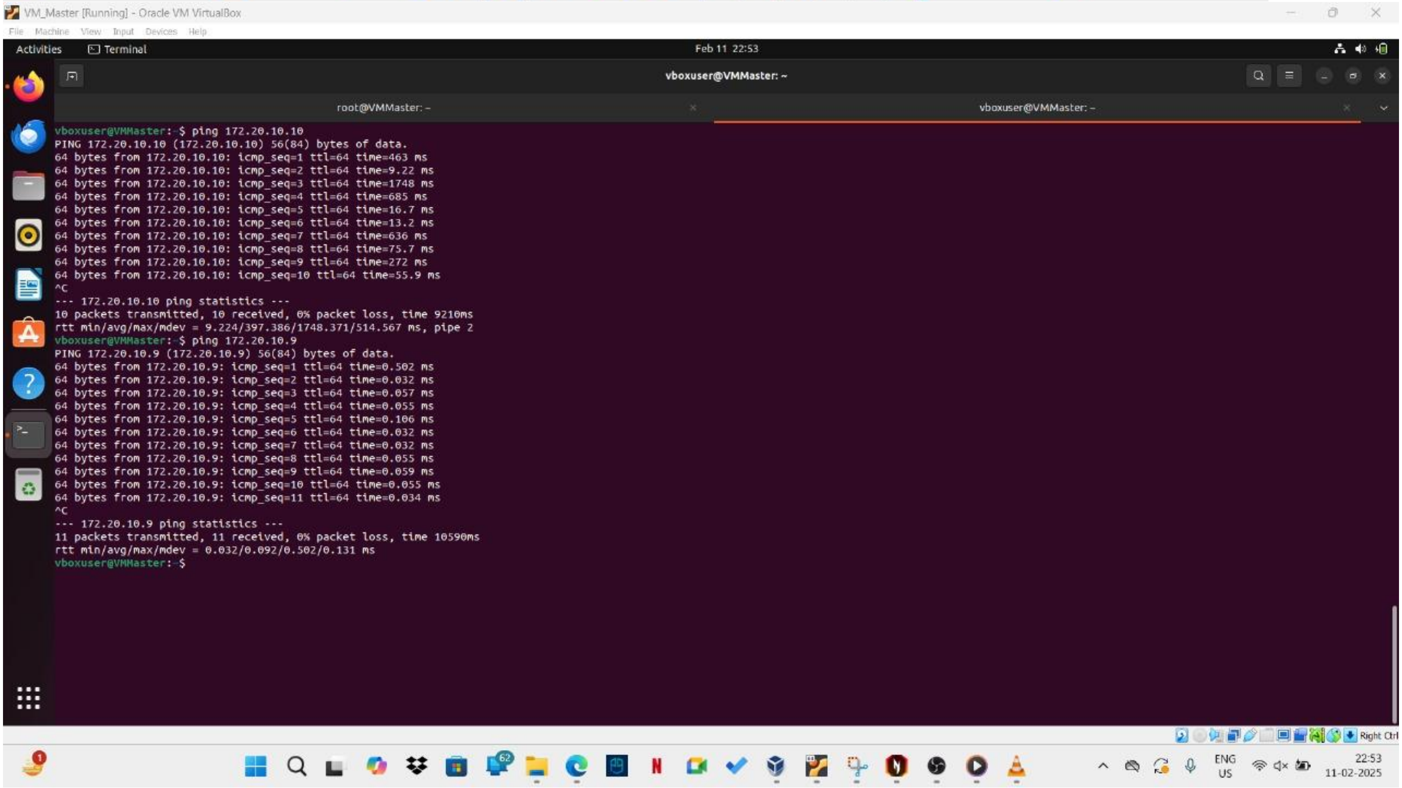
Conclusion:

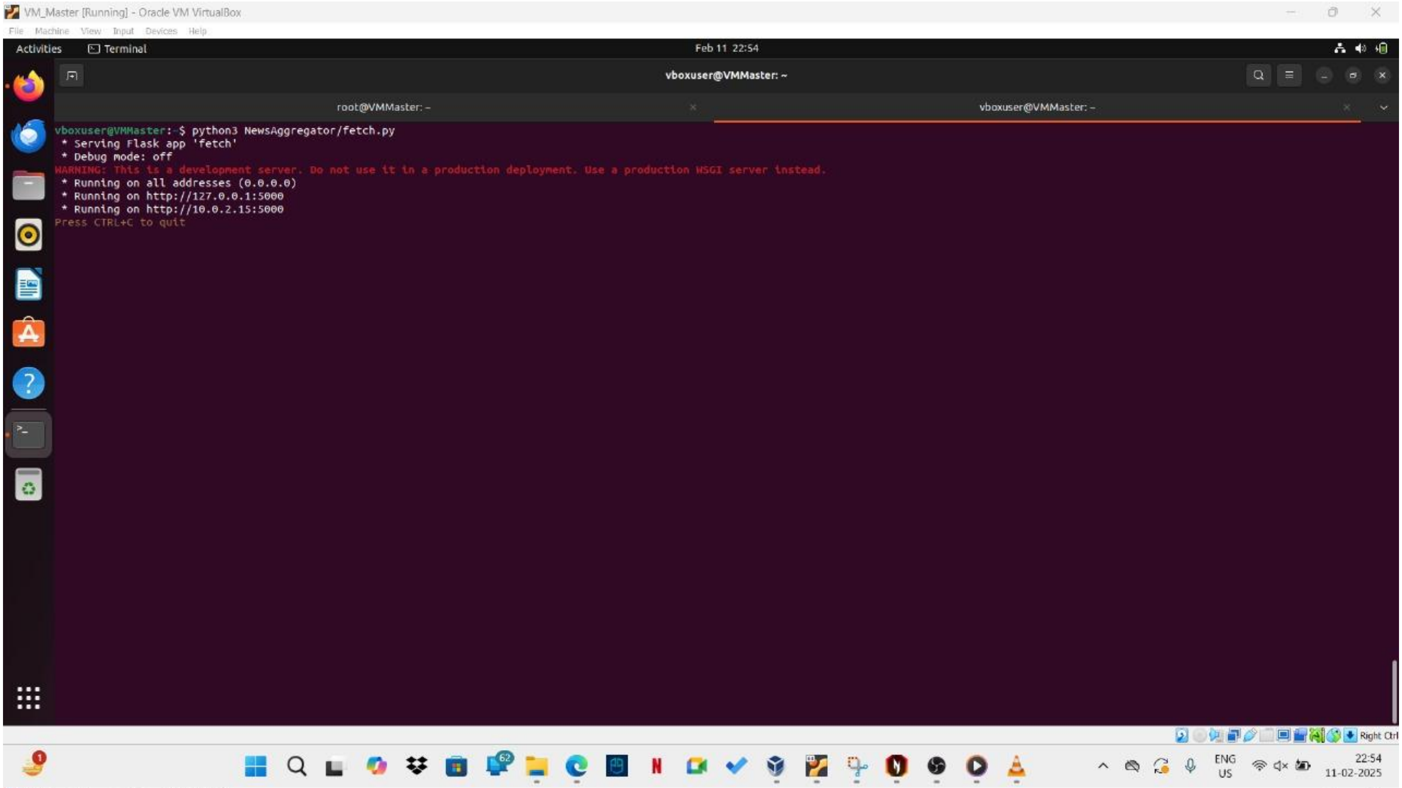
This report demonstrated the step-by-step process of creating multiple virtual machines, configuring network settings, deploying a Flask-based microservice, and enabling inter-VM communication. The successful execution of these steps confirms the correct deployment of the microservice application across multiple VMs in VirtualBox.

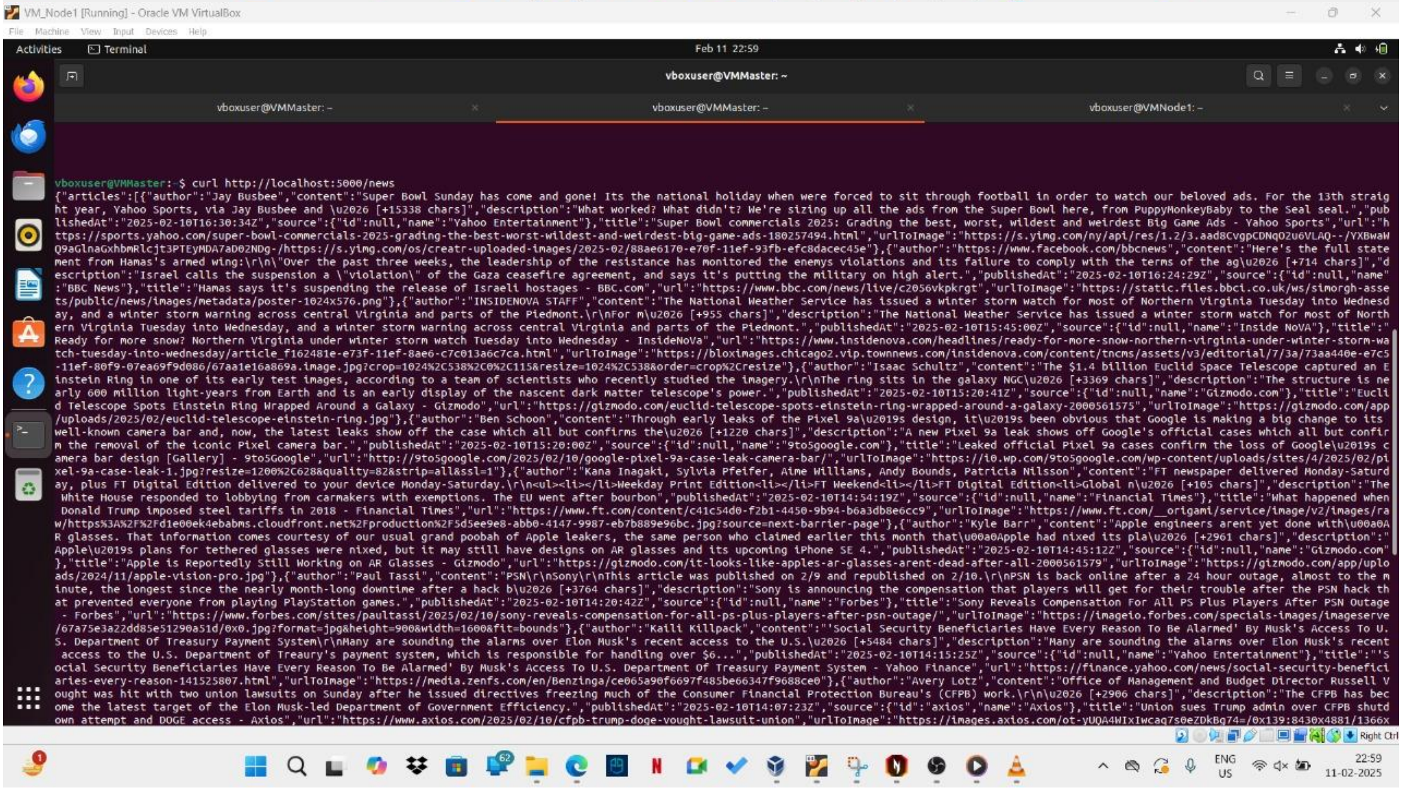

Oracle VM VirtualBox Manager
File Machine Help
Tools
VM_Master
Running
VM_Node1
Running
Node2
Running
New Add Settings Discard Show
General
Name: VM_Master
Operating System: Ubuntu (64-bit)
System
Base Memory: 3301 MB
Processors: 3
Boot Order: Hard Disk, Optical, Floppy
Acceleration: Nested Paging, KVM Paravirtualization
Preview
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: VM_Master.vdi (Normal, 32.08 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
Adapter 2: Intel PRO/1000 MT Desktop (Bridged Adapter, Intel(R) Wi-Fi 6 AX203)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None
ENG
US
22:49
11-02-2025


VM_Master [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities
Terminal
Feb 11 22:54
vboxuser@VMMaster: ~
root@VMMaster: ~
vboxuser@VMMaster: ~
vboxuser@VMMaster:~$ cat NewsAggregator/
Dockerfile fetch.py test.py
vboxuser@VMMaster:~$ cat NewsAggregator/fetch.py
import os
import requests
from flask import Flask, jsonify
app = Flask(__name__)
#NEWS_API_KEY = os.getenv("NEWS_API_KEY")
NEWS_API_KEY = "b636995e40a64a23b4896baeb0a8b71a"
@app.route("/news", methods=["GET"])
def get_news():
#if not NEWS_API_KEY:
#return jsonify({"error": "API key is missing"}), 500
url = "https://newsapi.org/v2/top-headlines?country=us&apiKey=b636995e40a64a23b4896baeb0a8b71a"
response = requests.get(url)
if response.status_code == 200:
return jsonify(response.json())
else:
return jsonify({"error": "Failed to fetch news"}), response.status_code
if __name__ == "__main__":
app.run(host="0.0.0.0", port=5000)
vboxuser@VMMaster:~$
Show Applications
Right Ctrl
ENG
US
22:54
11-02-2025


VM_Master [Running] - Oracle VM VirtualBox
Activities
Terminal
Feb 11 22:51
vboxuser@VMMaster: ~
root@VMMaster: ~
vboxuser@VMMaster: ~
vboxuser@VMMaster:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
ether 02:42:7d:b9:c4:ea txqueuelen 0 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::923f:61e5:15ca:e168 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:30:80:29 txqueuelen 1000 (Ethernet)
RX packets 5237 bytes 3748253 (3.7 MB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 3538 bytes 1126199 (1.1 MB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 172.20.10.9 netmask 255.255.255.240 broadcast 172.20.10.15
inet6 2409:40f0:1018:86ea:67fa:10f4:8cbb:699b prefixlen 64 scopeid 0x0<global>
inet6 2409:40f0:1018:86ea:25ed:4786:f4:6385 prefixlen 64 scopeid 0x0<global>
inet6 2409:40f0:1031:f55e:9db:89bf:5bfe:6987 prefixlen 64 scopeid 0x0<global>
inet6 fe80::835c:e06d:68c8:cc8d prefixlen 64 scopeid 0x20<link>
inet6 2409:40f0:1031:f55e:294a:9b32:1bb7:dc2b prefixlen 64 scopeid 0x0<global>
ether 08:00:27:c0:41:f4 txqueuelen 1000 (Ethernet)
RX packets 3556 bytes 473141 (473.1 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 3793 bytes 542700 (542.7 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 853 bytes 120440 (120.4 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 853 bytes 120440 (120.4 KB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
vboxuser@VMMaster:~$
ENG
US
22:51
11-02-2025


VM_Master [Running] - Oracle VM VirtualBox
Activities
Terminal
Feb 11 22:53
vboxuser@VMMaster: ~
root@VMMaster: ~
vboxuser@VMMaster: ~
vboxuser@VMMaster:~$ ping 172.20.10.10
PING 172.20.10.10 (172.20.10.10) 56(84) bytes of data.
64 bytes from 172.20.10.10: icmp_seq=1 ttl=64 time=463 ms
64 bytes from 172.20.10.10: icmp_seq=2 ttl=64 time=9.22 ms
64 bytes from 172.20.10.10: icmp_seq=3 ttl=64 time=1748 ms
64 bytes from 172.20.10.10: icmp_seq=4 ttl=64 time=685 ms
64 bytes from 172.20.10.10: icmp_seq=5 ttl=64 time=16.7 ms
64 bytes from 172.20.10.10: icmp_seq=6 ttl=64 time=13.2 ms
64 bytes from 172.20.10.10: icmp_seq=7 ttl=64 time=636 ms
64 bytes from 172.20.10.10: icmp_seq=8 ttl=64 time=75.7 ms
64 bytes from 172.20.10.10: icmp_seq=9 ttl=64 time=272 ms
64 bytes from 172.20.10.10: icmp_seq=10 ttl=64 time=55.9 ms
^C
--- 172.20.10.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9210ms
rtt min/avg/max/mdev = 9.224/397.386/1748.371/514.567 ms, pipe 2
vboxuser@VMMaster:~$ ping 172.20.10.9
PING 172.20.10.9 (172.20.10.9) 56(84) bytes of data.
64 bytes from 172.20.10.9: icmp_seq=1 ttl=64 time=0.502 ms
64 bytes from 172.20.10.9: icmp_seq=2 ttl=64 time=0.032 ms
64 bytes from 172.20.10.9: icmp_seq=3 ttl=64 time=0.057 ms
64 bytes from 172.20.10.9: icmp_seq=4 ttl=64 time=0.055 ms
64 bytes from 172.20.10.9: icmp_seq=5 ttl=64 time=0.106 ms
64 bytes from 172.20.10.9: icmp_seq=6 ttl=64 time=0.032 ms
64 bytes from 172.20.10.9: icmp_seq=7 ttl=64 time=0.032 ms
64 bytes from 172.20.10.9: icmp_seq=8 ttl=64 time=0.055 ms
64 bytes from 172.20.10.9: icmp_seq=9 ttl=64 time=0.059 ms
64 bytes from 172.20.10.9: icmp_seq=10 ttl=64 time=0.055 ms
64 bytes from 172.20.10.9: icmp_seq=11 ttl=64 time=0.034 ms
^C
--- 172.20.10.9 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10590ms
rtt min/avg/max/mdev = 0.032/0.092/0.502/0.131 ms
vboxuser@VMMaster:~$
ENG
US
22:53
11-02-2025


VM_Master [Running] - Oracle VM VirtualBox
Activities
Terminal
Feb 11 22:54
vboxuser@VMMaster: ~
root@VMMaster: ~
vboxuser@VMMaster: ~
vboxuser@VMMaster:~$ python3 NewsAggregator/fetch.py
* Serving Flask app 'fetch'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.0.2.15:5000
Press CTRL+C to quit
22:54
11-02-2025


VM_Node1 [Running] - Oracle VM VirtualBox
Activities
Terminal
Feb 11 22:59
vboxuser@VMMaster: ~
vboxuser@VMMaster: ~
vboxuser@VMMaster: ~
vboxuser@VMNode1: ~
vboxuser@VMMaster:~$ curl http://localhost:5000/news
22:59
11-02-2025